

CrewCut-QAOA

A quantum circuit that assigns storm repairs to two crews. Within 0.32% of the best plan from a hundred shots.

Team Briskit · Qollab x IonQ hackathon, October 2026

Summary

After a storm a utility decides which crew takes which repair. CrewCut answers that for two crews with QAOA, one qubit for each repair.

0.32%

from the best plan with a hundred shots, no plan scored beforehand. Random plans: 0.62%. Published algorithm: 1.47%.

One circuit layer · 20 feeder instances · mean · ahead of random plans on 20 of 20

0.09%

from the best plan with 200 scored plans, 150 of them to fit the cost. Random plans: 0.26%. Classical best first search: 0.00%.

Two layers · 30 feeder instances · median · means 0.16%, 0.31% and 0.11%

3 to 5x

fewer shots than standard QAOA to reach a best plan, on half the qubits.

6 to 10 independent repairs · two layers · median · ahead on every instance

52 vs 45%

of the gain kept at the error rates of an ion machine when half of the pair gates are removed.

12 independent repairs · two layers · 30 instances · median · noise model

Under each number: its setup. Every circuit is simulated; none ran on a machine. Best plan is defined on page 3.

After a storm, who fixes what first?

A utility has more broken lines than crews. A repaired line carries power only once the lines between it and the substation are repaired, so the order matters.

2,048

ways to split twelve repairs between two crews.

Twelve choices of crew A or crew B make 4,096 strings. Swapping the names of the crews gives the same plan, which leaves 2,048.

Each crew works its repairs in the order of a fixed rule. A plan is scored by the customer-hours without power, weighted by priority. **Best plan** means the best of the 2,048 splits; a schedule free to reorder each crew is 0.15% better on average.

Test case: the public IEEE 13 node feeder. Travel between sites is not modelled; Tan et al. fold it into the repair times. The 0.15% is over the 12 instances where the optimum is proven.

What a percent is worth

Weather outages cost the United States $18 to $33 billion a year. One percent of that is $180 to $330 million.



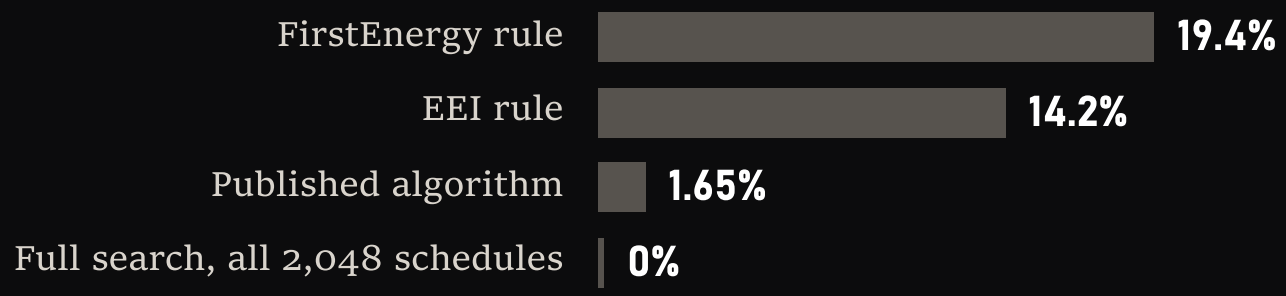
At that rate a tenth of a percent is $18 to $33 million, and the differences measured here, in dollars a year:

1.15 points	a hundred shots of CrewCut ahead of the published algorithm: $207 to $380 million.
0.30 points	ahead of a hundred random plans: $54 to $99 million.
0.09 points	classical best first search ahead of CrewCut at 200 scored plans: $16 to $30 million.

A conversion to show scale: it does not say that any utility runs these methods, and percentages from a 13 node test feeder are not shown to hold in a real storm. Cost is taken in proportion to outage time. The shots are a list price, without tuning; no run was made. Sources: docs/money in the repository.

The classical baseline

Every result is measured against classical methods. Their distance from the best plan:



The rules fix what brings back the most customers first (FirstEnergy), or the most customers in the least time (EEI). The published algorithm finds the best order for one crew, then hands the next job to whichever crew is free.

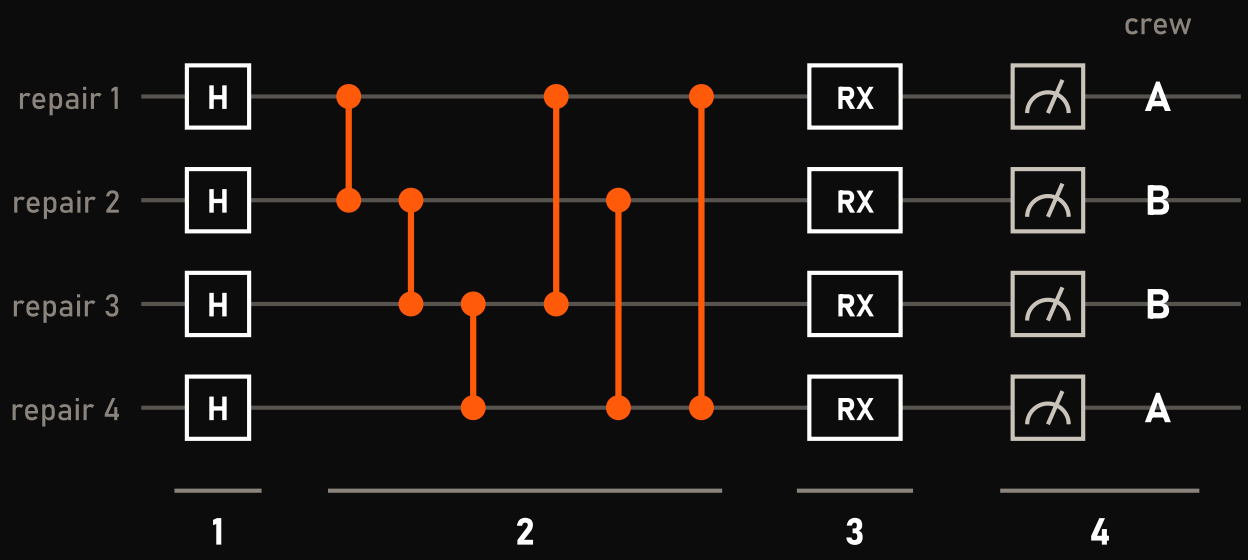
Why it is a fair baseline

1. **The answer is known.** Every schedule is scored, so each percent is a measured distance from the best plan.
2. **It is what is used and what is published.** The algorithm comes with a proof: never more than 1.5 times the best for two crews.
3. **Same budget.** Head to head, classical search and samplers score as many plans as the circuit does.

Mean over 300 instances; rules and algorithm from Tan et al., 2017. The published algorithm does well on most instances and badly on a few: its median here is 0.49%. On the 20 instances of page 9 its mean is 1.47%, on the 30 of page 11 it is 1.78%.

QAOA in one page

QAOA is a quantum algorithm for choosing among many options. One qubit holds one yes or no choice: here, crew A or crew B for one repair.



1. **Start** in an equal mix of every plan.
2. **Cost layer.** A two-qubit gate for each pair of repairs marks every plan by its cost.
3. **Mixer.** A rotation on each qubit turns those marks into odds, so cheap plans become likely.
4. **Measure.** Each run, a shot, returns one plan. Repeat and keep the best.

Four repairs shown. Steps 2 and 3 form one layer and can be repeated; each layer has two angles, tuned in simulation. Farhi, Goldstone and Gutmann, 2014.

From a plan to qubits

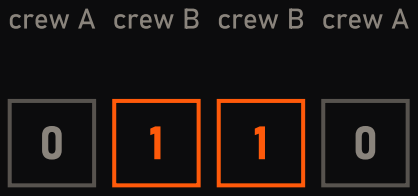
A plan is one choice per repair: crew A or B. A rule orders each crew, so only that choice needs qubits.

Standard QUBO



8 qubits. A plan only when each column holds exactly one 1.

CrewCut



4 qubits. Every string of bits is a plan.

$$\text{cost} = \Sigma_j\, w_j\, p_j + \Sigma_{i \text{ before } j, \text{ same crew}}\, w_j\, p_i$$

w is the priority weight of a repair, p its hours. Exact when repairs do not depend on each other; for a feeder see page 8.

Standard QUBO. The pair term $w_j p_i\, x_i x_j$ for two bits of one crew, plus a penalty $A(1 - x_A - x_B)^2$ on each repair so that it lands on exactly one crew.

CrewCut. Two repairs pay their pair term when their bits are equal: $\tfrac{1}{2}\, w_j p_i\, Z_i Z_j$ plus a constant. The cheapest plan cuts the heaviest pairs apart, which is weighted MaxCut. One two-qubit gate for each pair, no penalty.

Both forms use the same order rule: highest weight per hour first (Smith's ratio rule); on a feeder, the best order for one crew. One bit form: Skutella, 2000.

What CrewCut does differently

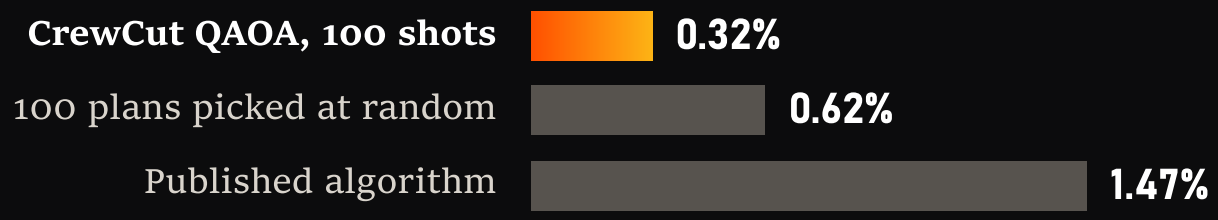
12, not 24	qubits for twelve repairs, and 66 two-qubit gates in a layer in place of 144 (205 against 420 on a heavy hex chip).
No penalty	Nothing to tune, and every shot is a valid plan.
Order by rule	The sequence inside a crew costs no qubits.
Fitted pairs	On a feeder a line waits for the lines upstream of it, so the cost is not exactly a sum over pairs. The pair weights are fitted to 150 scored plans by regression. The best plan of the pair cost is then 2.05% above the best plan, against 16.9% unfitted.
Half the gates	The weakest half of the pair terms can be dropped: 33 gates in a layer in place of 66, and more of the gain survives noise (page 13).
What is new	The one bit form is from scheduling theory (Skutella, 2000). New here: its use for restoration on a feeder, the fitted pairs, the thinned layer, and the test against standard QAOA and classical search.

Gate counts with any two qubits able to interact, as on trapped ions; heavy hex is a 115 qubit layout. Fit figures: mean over 20 feeder instances.

Where quantum helps

0.32%

from the best plan, with a hundred shots and no plan scored beforehand.



Ahead of a hundred random plans on 20 of 20 instances and of the published algorithm on 13 of 20; on 5 of the other 7 that algorithm already has the best plan. Medians: 0.29% against 0.65% and 0.42%.

Test feeder, 12 repairs, two crews, 20 instances, means, ideal simulator. One circuit layer on the pair cost of page 7, unfitted: 98% of the variance of the true cost over all plans sits in pair terms. With this unfitted cost more layers score worse (0.66% and 1.13% at two and three). With the fitted cost of page 11, two layers are best.

A twentieth of the scoring

A full search scores all 2,048 schedules. CrewCut scores a hundred.



100 of 2,048, and it lands 0.32% from the plan the full search finds.

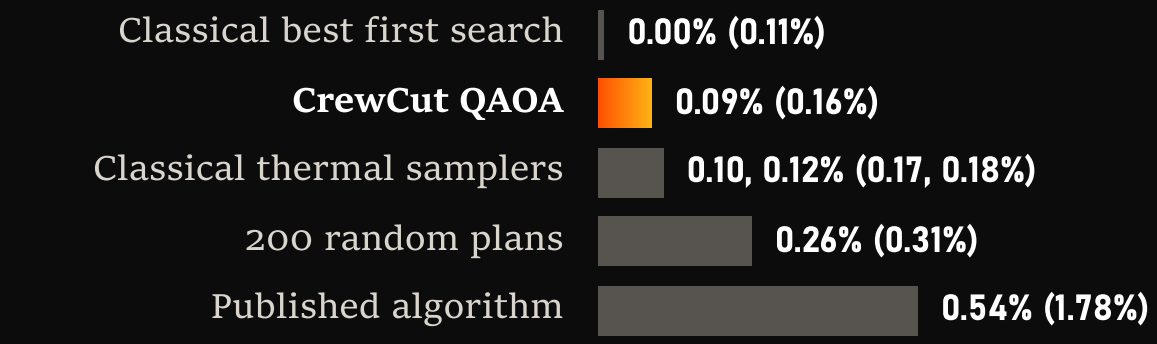
The circuit has two angles to tune. Tuned on one instance and reused on the other nineteen, they give 0.29% to 0.35%, against 0.32% with angles tuned for each. Tuning is done once, not for every storm.

Each shot is 2.6 times as likely to be a best plan as a random pick. Lit dots are placed at random. Scoring is the step that grows with the network. At twelve repairs a full search is itself quick (page 15), and tuning the angles in simulation takes longer than that search.

Virtually the best plan

0.09%

from the plan a full classical search finds, after scoring a tenth of the schedules.

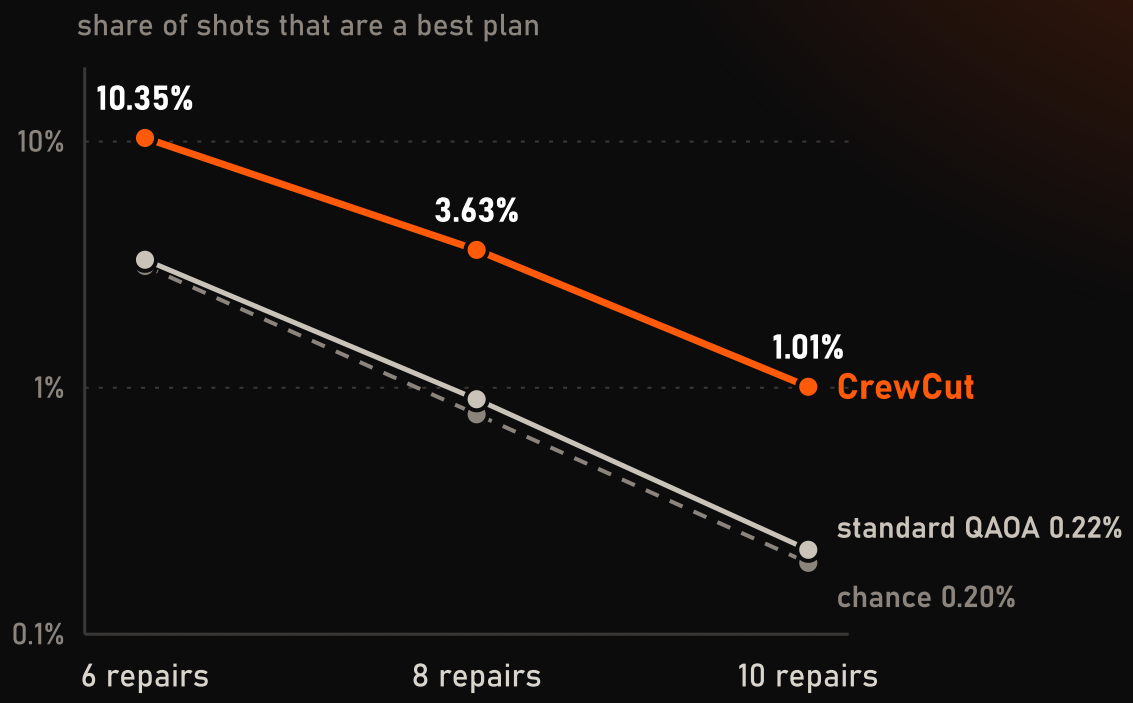


All get 200 scored plans. For the fitted methods the first 150 are random plans that fit the cost, and the last 50 differ. Fifty circuit shots take CrewCut from 0.25% to 0.09%; fifty more random plans take the random row from 0.34% to 0.26%.

Best first search scores plans in the order of the fitted cost and stays ahead: it reaches the best plan on 16 of 30 instances, CrewCut on 8. The thermal samplers draw plans by fitted cost at two temperatures; CrewCut is level with them.

30 feeder instances, two circuit layers; medians, with means in brackets. The two rows start apart at 150 by chance: each draws its own random plans. The mean of the published algorithm is in line with the 1.47% and 1.65% of pages 9 and 5.

Ahead of standard QAOA

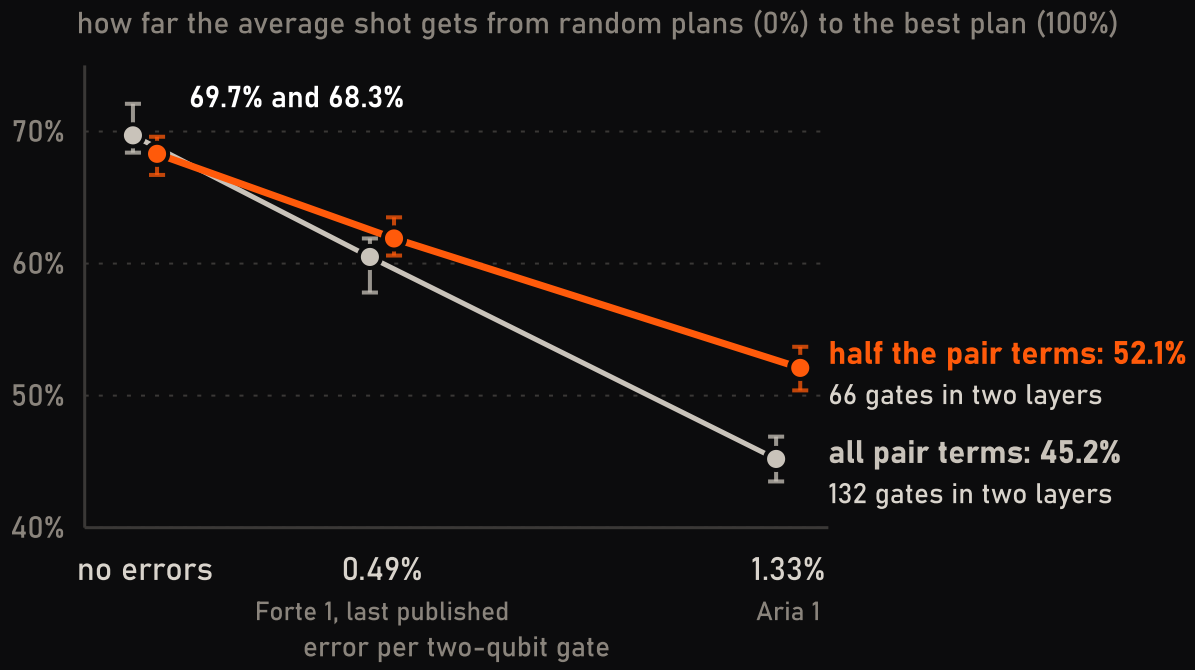


3 to 5 times as many best plans, on every instance, and the gap widens with size.

Standard QAOA is the textbook penalty form with the same order rule, the same trainer and six starts. Its penalty is run at up to eight settings and the best is picked for each instance, using the answer. It still stays near the rate of chance.

Two layers, ideal simulator, medians; 30 instances at 6 and 8 repairs, 10 at 10. Shots for a 95% chance of a best plan: 27 against 89, 81 against 330, 296 against 1,372. Independent repairs, where both forms encode the same exact cost. The standard form needs 20 qubits at 10 repairs and that run took three hours, so the comparison stops there. Chance is 2 in 2^n: a plan and its mirror.

Made for noisy machines



With half the pair gates removed, the circuit keeps more of its gain at the higher error rate.

Without errors the full circuit is slightly ahead. At 1.33% error per two-qubit gate the thinned one leads, 52.1% against 45.2%, and the quartiles do not overlap.

Hosted on Qollab (one instance, 2,000 shots a run), shots that are a best plan: 15 with half the terms against 2 with all on the IonQ Forte 1 model, 9 against 6 on the IBM Fez model. Chance gives 2.

12 independent repairs, two layers, 30 instances, medians with quartiles, at the error rates IonQ documents. Forte 1: the last published rates; IonQ updated that model in September 2025 and has not yet documented it. Three error rates and single hosted runs are an indication, not a law.

Beyond the hackathon

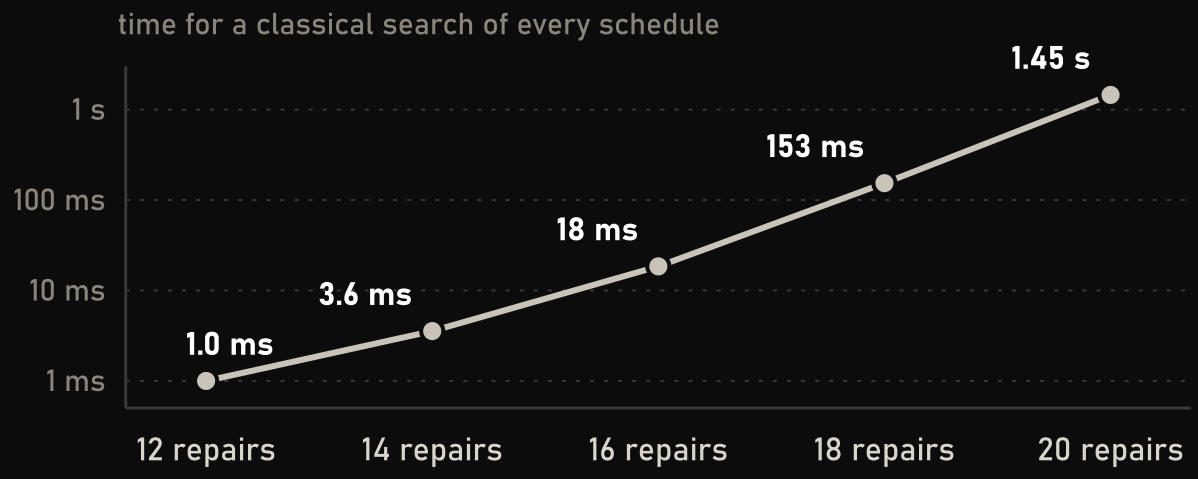
Not a claim that QAOA should restore power today. Four groups can use what was measured.

Research	A measured price for the penalty: its valid shots get 0.1% to 7.8% of the way from random plans to the best plan, those of CrewCut 56% to 73%. Open: which other problems of two resources have a pair form without a penalty.
Education	No quantum background is needed: two crews, repair hours, priorities. It runs at 6 to 20 qubits on a laptop and on the free simulators of Qollab. A syllabus names the source behind each file.
Industry	Utility rules sit 14% and 19% from the best plan, and a classical algorithm closes most of that. Near use: a small test workload with a known answer for compilers and machines. One percent of a year of weather outages is $180 to $330 million. Same form of problem: flight delays ($32.9 billion, 2007), operating rooms ($37.45 a minute).
Community	Restoration order decides whose power returns first, so a claim about it should be checkable by anyone: results tied to result files, classical wins printed beside quantum ones, MIT licence.

Research figures: 6 to 10 independent repairs, medians, the penalty form at its best penalty. Nothing was run for flights or operating rooms. Dollar sources: docs/money in the repository.

What quantum advantage would take

Quantum advantage means beating the best classical method. At twelve repairs a classical search of every schedule takes about a millisecond, so there is none to claim.



The schedules double with each repair and the search time follows: 1,450 times as long at 20 repairs as at 12. Past the reach of a full search the classical options are rules, samplers and local search.

Advantage would take a quantum sampler that beats those at such a size. That is not measured here. In place for it: a circuit of one qubit per repair, under half the gates of standard QAOA, and a thinned cost layer that keeps more of its gain under noise.

Search over independent repairs on one laptop thread, median of 3 runs. Tuning one layer of the circuit in simulation takes 76 ms at 12 repairs.

Bigger problems, more crews

One more repair is one more qubit. The list of schedules doubles.

		CrewCut		standard QAOA	
repairs	schedules	qubits	pair gates	qubits	pair gates
12	2,048	12	66	24	144
20	524,288	20	190	40	400
30	537 million	30	435	60	900
40	550 billion	40	780	80	1,600

More crews. One bit per repair is the two crew case. Four crews take two bits per repair, still with no penalty, but two repairs then share a crew only when both bits agree, which takes gates on four qubits. Splitting the crews in halves, one CrewCut at a time, is the other route. Neither is built here.

Other networks. On nine flow networks with loops the question is which links to restore. With 200 scored plans classical best first search ends 0.5% from the best on average, the circuit 2.1%, random plans 2.6%.

Pair gates are for one layer. Rows above 12 repairs are counts from the circuit definitions; nothing is run at those sizes.

Method and scope

Instances are drawn on the IEEE 13 node test feeder: 12 damaged lines, two crews, priority weights and repair times of 1 to 10 hours drawn at random. Every plan is scored, so the best one is known and each figure is a distance from it.

The repository holds more than 30 experiment scripts with saved results and more than 80 checks. Each check passes on the code and fails on a deliberately broken copy.

Scope. Two crews. No travel time between sites. Every circuit is simulated, with noise models where stated; nothing ran on a machine. Classical search is exact and quick at this size, and classical best first search is ahead of CrewCut by 0.09 points at 200 scored plans (pages 11 and 15). Without a fit, feeder results use one circuit layer and deeper circuits score worse; with the fit, two layers are best. The comparisons with standard QAOA, under noise and on Qollab use independent repairs. Dollar figures convert percentage points of a test feeder at national rates to show scale. The spread of page 9 and the reuse of angles of page 10 come from a rerun of experiments/crew_assignment.py with output for each instance.

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github.com/NikoKleion/CrewCut-QAOA-Team-Briskit-
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